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Introduction

Microfluidics is the study of the science and technology of systems that use small amounts of fluids using channels in the microscale [1]. Currently, the technology has seen major application in biomedical fields such as diagnostic capabilities [2]. The objective of this experiment is to gain a better understanding and verify the governing equations, *If otherwise stated, variables are defined equivalently in the previous equation*, this includes:

Conservation of mass: $\rho US = \text{constant}$ [kg/s], where ρ [kg/m^3] is the density of the fluid, U [m/s] is the velocity, S [m^2] is the cross sectional area. This equation shows that mass flow rate is constant under steady state. [1]

Bernoulli's Equation: $P + \frac{1}{2}\rho U^2 + \rho gz = \text{constant}$ [N/m^2], where P [N/m^2] is the pressure, $g = 9.81 m/s^2$, z [m] is the height of fluid, this is a result of the conservation of energy for an incompressible, non-viscous flow under steady state. [2a]

And using the following equations to determine the velocity profile of the flow:

Hagen-Poiseuille Equation: $Q = \frac{S^2}{8\pi\mu L} (\Delta P + \rho g \Delta z)$, where $S = \pi(\text{radius})^2$ [m^2] is the surface area, μ [p] is the dynamic viscosity, L [m] is the length of the pipe section, [3]

Flow Rate Equation: $Q = SU_{mean}$, where U_{mean} [m/s] is the **mean** velocity [4]

Experimental Procedures

In this experiment, a microfluidics chip with different channels is placed on an inverted microscope in a petri dish, the chip is connected to a fluid source syringe and a waste reservoir tube via tube channels. The source syringe is placed at a height z relative to the chip, generating the fluid flow via gravitational forces, thus flow speed can be manipulated via changing z . Images are produced in the LAS software containing streaks of fluid fluorescent seeding, thus:

Fluid Velocity: $U = d_a/t$, where U [m/s] is the velocity of the particle, d_a [m] is the physical length of the streak in the image, t [s] is the exposure time read in software [5]

To determine the physical length of the streak, a pixel to actual length ratio has to be considered (given by the LAS software), this can be expressed as:

Actual length: $d_a = d_p(r)$, where d_p [m] is the pixels the streak spans, r [$m/pixels$] is the pixel to length ratio. [6]

The experiment consists of a.) Measuring velocity profiles of different sections, b.) Changing the height z to produce different flow speeds. *Please refer to the lab manual for more details* [3]

Error Analysis

One source of quantifiable error in this experiment is the various measurement uncertainties. Note that all pixel measurements are made using an online tool¹, with an estimated pixel uncertainty of ± 2 **pixels** to account for each pixel edge. To determine the uncertainty of the ratio,

¹https://hayson.me/My_Website/microfluidicthing/

r in Eq 6, take $\delta d_{p,scale} = 2 \text{ pixels}$; the instrumental uncertainty of the scale length, $\delta d_{a,scale} = 5 \text{ }\mu\text{m}$ from the LAS software. Thus,

$$\delta r = r \sqrt{\left(\frac{\delta d_{p,scale}}{d_{p,scale}}\right)^2 + \left(\frac{\delta d_{a,scale}}{d_{a,scale}}\right)^2} = r \sqrt{\left(\frac{2}{d_{p,scale}}\right)^2 + \left(\frac{5}{d_{a,scale}}\right)^2} \text{ for Eq 5} \quad [7]$$

Since images are 2 dimensional, δd_p in Eq 6 is propagated and estimated as $\pm 2.8 \text{ pixels}$, thus,

$$\delta d_a = d_a \sqrt{\left(\frac{\delta r}{r}\right)^2 + \left(\frac{\delta d_p}{d_p}\right)^2} = d_a \sqrt{\left(\frac{\delta r}{r}\right)^2 + \left(\frac{2.8}{d_p}\right)^2} \text{ for Eq 6} \quad [8]$$

And from the LAS software, instrumental uncertainty of time δt in Eq 5 = $\pm 0.05 \text{ ms}$, thus,

$$\delta U = U \sqrt{\left(\frac{\delta t}{t}\right)^2 + \left(\frac{\delta d_a}{d_a}\right)^2} \text{ for Eq 5, } \delta U_{mean} = \left(\sum_i (\delta U_i)^{-2}\right)^{-1/2} \text{ for Eq 4} \quad [9]$$

Another quantifiable error arises from the imperfections of the chip design, i.e. ridges in the chip.



Figure 1a (left) shows a ridge under 20x magnification; Figure 1b (right) 5x magnification. The maximum extension of ridges in the above section can be quantified to obtain $\sim 20 \text{ }\mu\text{m}$. These ridges allow small eddies to form along the wall, increasing frictional forces. Hence, these ridges “compresses” the velocity profile due to a reduced surface area, the mean velocity also decreases. Finally, the pressure drop increases due to the increased frictional forces, as pressure change $\propto$ frictional forces according to the Darcy-Weisbach Equation in turbulent flow [4].

Non-quantifiable errors include the assumption of Bernoulli’s Equation in areas of higher velocity and small air bubbles present in the channels; these errors can be selectively eliminated by preventing the collection of data in nearby regions. In general, quantifiable errors are more significant than non-quantifiable errors in this experiment.

Results and Discussion

Velocity Profile in Straight Channel

Refer to Figure 2, the flow is laminar in the straight channel due to its parallel, layered and straight movement, as compared to the chaotic and unpredictable turbulent flow.

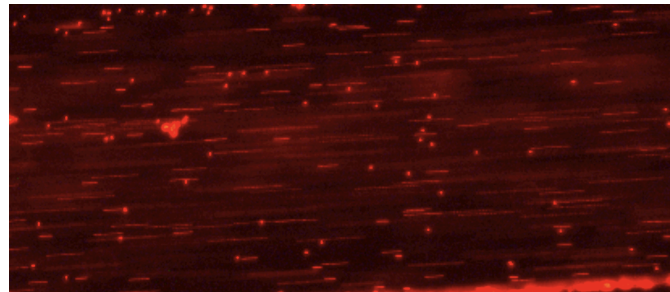


Figure 2, flow in the straight channel, note the parallel and layered arrangement of streaks, also note that shorter streaks are observable near the boundary (top or bottom).

Eq 3 can be applied to model this viscous flow, where the velocity profile is given as:

$$U(r) = \left(-\frac{dP}{dx} \right) \left(\frac{R^2 - r^2}{4\mu} \right) = k(R^2 - r^2), k \text{ is some constant,} \quad [10]$$

where dP/dx [Pa/m] is the pressure gradient across flow direction, μ [p] is the dynamic viscosity, R [m] is the radius of the tube, r [m] is the distance from the center of the tube.

Eq 10 models the velocity profile as a parabolic shape with the maximum velocity at the middle, and minimum 0 velocity at the sides, which agrees with the theoretical idea of the no-slip condition being present. Velocities and distances from the radius are then collected and plotted against Eq 10, they are plotted below, where measured $R = -114.7 \pm 6.2 \mu\text{m}$:

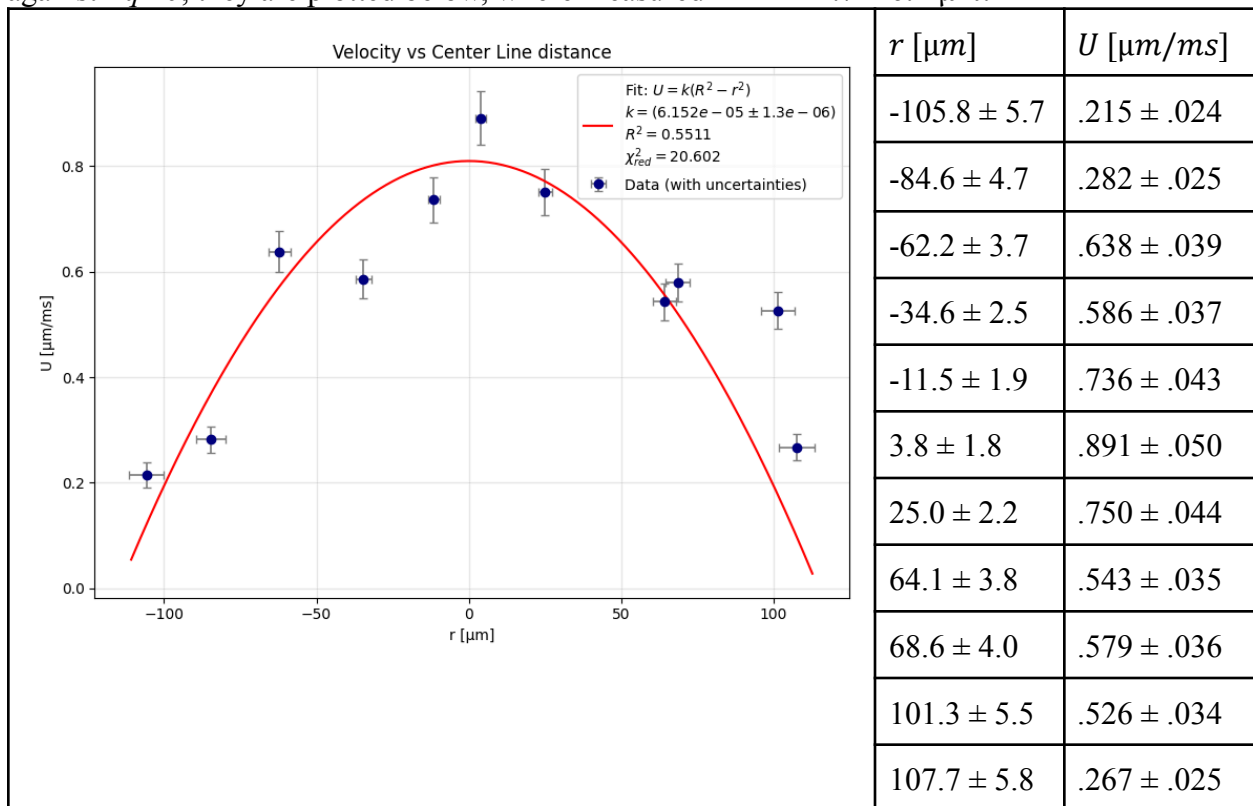


Figure 3a (left), best fit of collected data, note the data points do not coincide well with the fit.

Table 3b (right), Data collected with uncertainties.

From the plot and data, a curved trend is observed, with velocity reaching maximum at the middle, and dropping off near the sides. However, since the R^2 value $\ll 1$ and $\chi^2_{\text{red}} \gg 1$, it indicates a not so accurate fit with underestimated uncertainties, this is likely due to the difficulty of getting accurate velocities from the streaks, thus no strong conclusions can be drawn.

Height and Velocity in Straight Channel

Refer to Eq 2a, when considering a straight channel, since z is constant, thus, the velocity would increase when the pressure decreases, vice versa. Furthermore, increasing the height of the syringe increases the velocity of the straight channel by increasing the constant (see Appendix). Refer to Eq 3,4 and the ideal setup in the lab manual [3], the following relation is defined:

$$\rho g(z_3 - z_1) = 8\pi\mu((L_{2,3} + L_{3,4} + L_{5,6})/S_2^2 + (L_{4,5})/S_3^2)U_3, \text{ where } L_{i,i+1} [m] \text{ is the distance between sections, } S_i [m^2] \text{ is the cross-sectional area, } U_3 [m/s] \text{ is the straight channel velocity, } z_1 [m] \text{ is the syringe height} \quad [11]$$

This shows a linear relation between z_1 (syringe height, $\delta z = \pm 0.5 \text{ mm}$, from instrument error) and U_3 (mean exit velocity) is expressed in Eq 12, and plotted below:

$$U_3(z_1) = \alpha z_1 + \beta, \text{ where } \alpha [1/s] \text{ and } \beta [m/s] \text{ are fit parameters,} \quad [12]$$

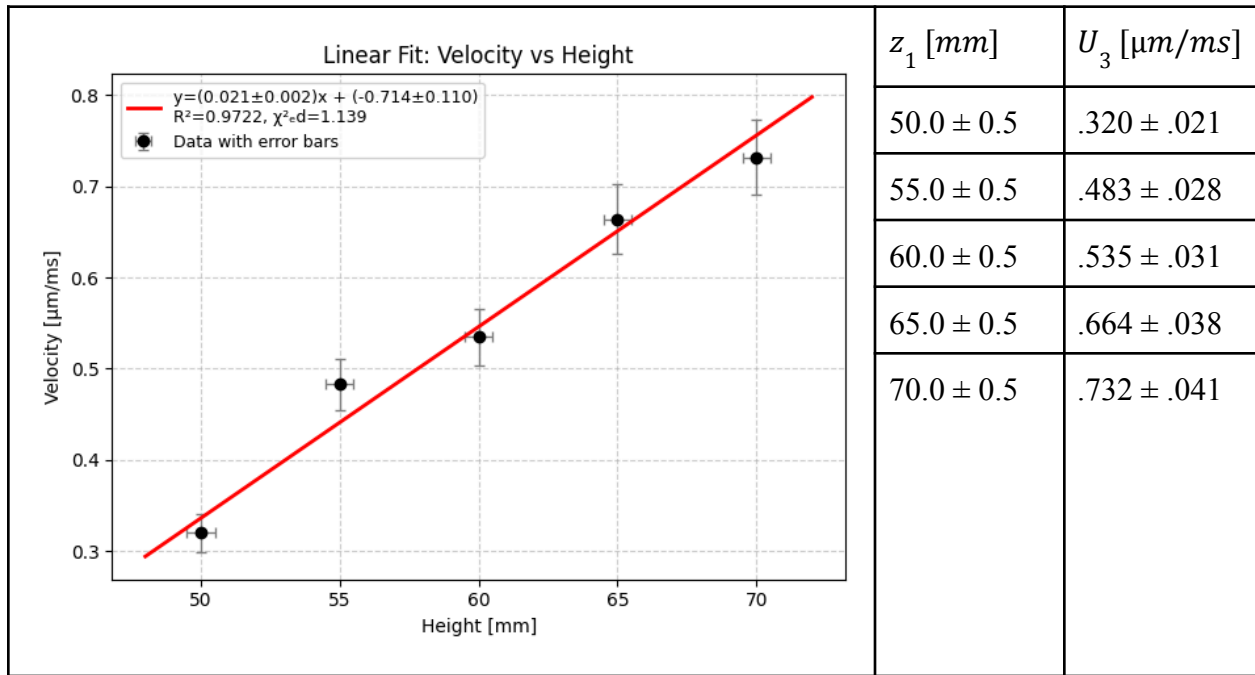


Figure 4a (left), best fit of collected data, note the data points **do** coincide well with the fit.

Table 4b (right), Data collected with uncertainties.

The collected data shows a good fit according to the fit parameters, and agrees with the theoretical formulation in Eq 11, indicating that Hagen-Poiseuille Principles are applicable in this case, correctly indicating an increasing trend.

Channels of Different Size

Below are the images obtained:

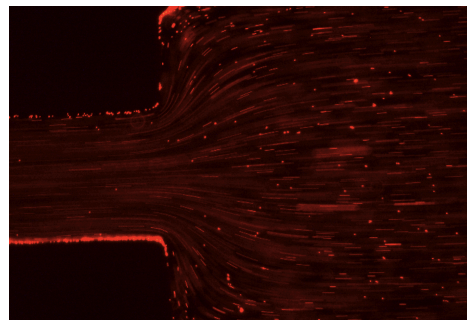
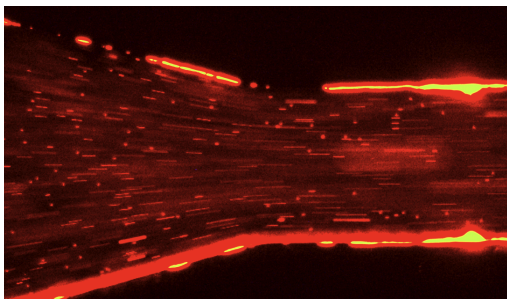


Figure 5a (left), Gradual change, note the smooth transition and overall parallel layers
Figure 5b (right), Abrupt change, note the curved layers and chaotic flow in the corners.

Assuming that the flow is steady and incompressible (*constant* ρ), also noting that in a Poiseuille flow, $U_{mean} = U_{max}$ (or $U_{centerline}$) / 2 [3], these relations can be combined with Eq 1 to give:

$$\frac{U_{2,mean}}{U_{1,mean}} = \frac{U_{2,max}}{U_{1,max}} = \frac{S_1}{S_2} = \left(\frac{r_1}{r_2}\right)^2, \text{ since } S = \pi r^2, \text{ and uncertainties can be propagated as,} \quad [13]$$

$$\delta\left(\frac{U_{2,mean}}{U_{1,mean}}\right) = \frac{U_{2,mean}}{U_{1,mean}} \sqrt{\left(\frac{\delta U_{2,mean}}{U_{2,mean}}\right)^2 + \left(\frac{\delta U_{1,mean}}{U_{1,mean}}\right)^2}, \delta\left(\frac{r_1}{r_2}\right)^2 = 2\left(\frac{r_1}{r_2}\right) \sqrt{\left(\frac{\delta r_1}{r_1}\right)^2 + \left(\frac{\delta r_2}{r_2}\right)^2} \quad [14]$$

Gradual Change (Larger $S_1 \rightarrow$ Smaller S_2)		Abrupt change (Smaller $S_1 \rightarrow$ Larger S_2)	
$U_{2,mean}/U_{1,mean}$	$(r_1/r_2)^2$	$U_{2,mean}/U_{1,mean}$	$(r_1/r_2)^2$
2.14 ± 0.19	2.50 ± 0.25	0.17 ± 0.03	0.14 ± 0.02

Table 6, experimental data summarized

Overall, the data agrees with the theoretical formulation within 2 experimental uncertainties, a weak agreement can be established. Ridges are clearly present and serve as a source of error, besides viscous forces in the abrupt change also causes errors. Observe that the gradual change allows parallel, layered and straight pathline; an abrupt change shows a curved path instead, less defined pathlines are visible near the corner of the change as well. In the straight channel, the flow is non-turbulent due to its straight path and predictability; In the abrupt channel, small eddies and curved pathlines contribute to the unpredictability and hence the turbulent flow. In daily applications, flow transitions are commonly found due to its capability of manipulating flow speed and properties and the physical constraints of the flow channel. In general, a smooth transition is preferred as less unpredictability is involved and lesser energy loss is present [5].

Channels with Bends

Below are the images obtained:

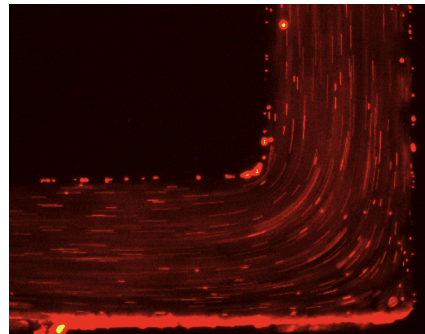
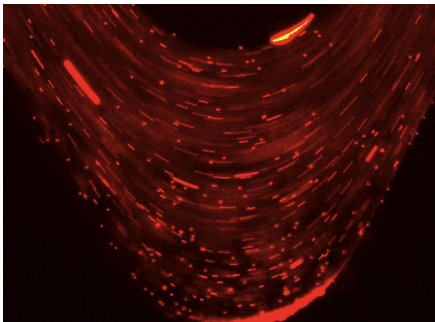


Figure 7a (left), Smooth bend, note the overall gradual change and conserved velocities.

Figure 7b (right), Sharp bend, note the less defined, swirls at the corner of the bend.

Assuming the same properties of the previous flow, in addition with equal channel widths, Eq 13 can be simplified into: $U_{2,mean} = U_{1,mean}$. Experimental data is summarized below:

Smooth Bend Velocity, U_c [$\mu\text{m/ms}$]		Sharp Bend Velocity, U_c [$\mu\text{m/ms}$]	
$U_{1,c}$	$U_{2,c}$	$U_{1,c}$	$U_{2,c}$
1.72 ± 0.08	1.64 ± 0.07 (95.3%)	1.38 ± 0.07	1.21 ± 0.06 (87.7%)

Table 8, experimental data summarized.

Overall, the velocity for the smooth bend agrees within 2 SDs, showing a weak equal/slight drop relation; the velocities for the sharp turn falls out of 2 SDs, indicating a decreasing relation. This decreasing relationship arises from the energy loss from ridges and the curved motion that the theory did not account for; sharp bend loses more kinetic energy by collisions with the tube.

The flow in both cases follows the tube geometry in both cases, with a more dense flow closer to the convex corner. The smooth bend has a flow that fills the entire space, while the sharp bend has flow that ignores the concave corner, this is due to the immense pressure gradient to overcome. In addition, the width of the stream tube decreases as the flow is closer to the convex corner. This is hypothesized to be the result of the interaction between the centripetal acceleration and the no-slip condition. Furthermore, the slight swirls and eddies in the corner of the sharp bend indicates a turbulent flow.

Conclusions

In conclusion, a weak parabolic relation can be observed in the velocity profile; this agrees weekly with the Hagen-Poiseuille velocity profile. On the other hand, the height and velocity relation agrees strongly with the Hagen-Poiseuille Relation. Similarly, the velocity and area ratio shows moderate agreement in the varying width; the velocity shows a slight decrease in the bending channel, both showing agreement with the massflow ratio. In conclusion, fundamental equations are shown and proven in varying degrees of agreement.

Bibliography

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Appendix

Analysis with Bernoulli's Principle

Had Eq 2 been applied instead, it can be rearranged to give:

$$\text{Rearranged form: } \frac{1}{2}\rho U_{n+1}^2 = \sqrt{\Delta P + \rho g \Delta z + \frac{1}{2}\rho U_n^2} [m/s], \quad [2b]$$

where $\Delta P [N/m^2]$ is the pressure change from point n to n+1, $g = 9.81 m/s^2$ is the gravitational constant, $\Delta z [m]$ is the height change of fluid from point n to n+1 *other variables are defined the same throughout the report.*

In addition, Torricelli's Principle gives that,

$$U_{n+1} = \sqrt{2g\Delta z} [m/s], \text{ all variables are defined the same throughout the report.} \quad [\alpha]$$

Combining both equations, the following is obtained:

$$U_3 = \frac{2}{\rho} \sqrt{(P_3 - P_2) + \rho g(z_1 - z_3)} [m/s], \quad [\beta]$$

all variables are defined the same throughout the report.

This is then fitted into the equation and summarized below:

$$U_3(z_1) = m\sqrt{z_1 + b}, \text{ where } m [m^{0.5}/s] \text{ and } b [m/s] \text{ are fit parameters} \quad [\gamma]$$

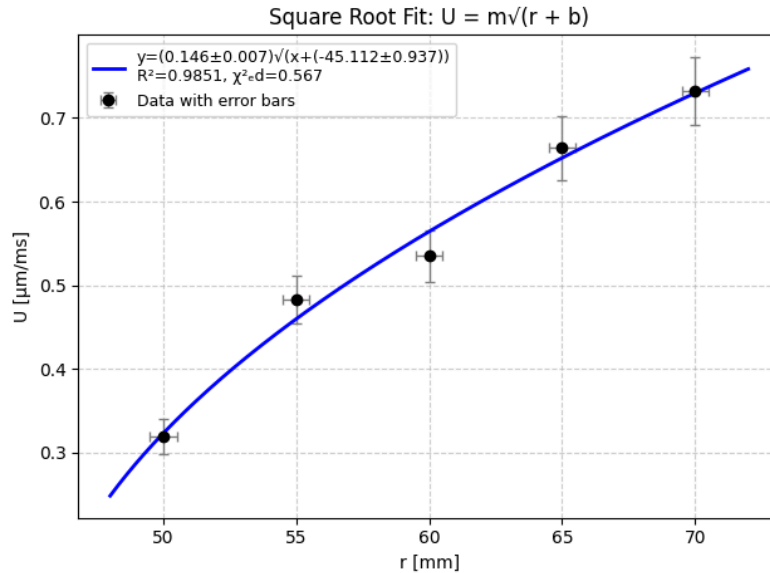


Figure X, best fit of collected data, note the data points **do** coincide well with the fit.

Again, the collected data shows a good fit according to the fit parameters, and agrees with the theoretical formulation in Eq γ , indicating that Bernoulli's Principles are equally applicable in this case, correctly indicating an increasing trend.